

House Price Prediction – Project Summary

Overview

This project focused on developing machine learning models to predict residential property prices using various house-related features such as area, number of bedrooms, bathrooms, parking availability, furnishing status, and location preferences. The objective was to assist buyers, sellers, and real estate businesses in estimating property values more accurately using data-driven methods rather than relying solely on market intuition.

Data Preparation and Exploration

The Housing Prices Dataset was loaded and explored using Pandas. Initial analysis included examining dataset dimensions, identifying the target variable (price), checking for missing values, and reviewing feature distributions. Duplicate records were removed, and categorical variables such as furnishing status, air conditioning, guest room availability, and preferred area were converted into numerical representations using one-hot encoding. This preprocessing ensured that the dataset was suitable for machine learning algorithms.

Model Development

Two regression models were trained and evaluated:

1. **Linear Regression**
2. **Random Forest Regressor**

The dataset was split into training (80%) and testing (20%) subsets to assess how well the models generalized to unseen data. Model performance was evaluated using standard regression metrics, with particular emphasis on the R^2 score.

Results

Model	R^2 Score
Linear Regression	0.6529
Random Forest Regressor	0.6133

The Linear Regression model achieved the highest R^2 score of approximately **0.65**, meaning it was able to explain around **65% of the variation in house prices** based on the available features. Surprisingly, the Random Forest model performed slightly worse, indicating that the relationships within this dataset are relatively linear and can be effectively captured by a simpler regression approach.

Key Findings

The analysis showed that property area, number of bathrooms, parking availability, air conditioning, and location-related features were among the strongest contributors to house prices. Larger homes with better amenities consistently commanded higher prices. The correlation analysis also revealed a strong positive relationship between area and price, confirming that house size remains one of the most influential pricing factors.

One unexpected finding was that the more complex Random Forest model did not outperform Linear Regression. This suggests that the dataset does not contain highly complex non-linear patterns and that simpler models may be sufficient for reliable price estimation.

Conclusion and Recommendation

Overall, the project demonstrated that machine learning can provide useful insights into housing market trends and property valuation. While the models are not perfect, the best-performing model explains a substantial portion of price variation and can serve as a practical decision-support tool. For real estate businesses, focusing on properties with larger living areas, multiple bathrooms, ample parking, and desirable locations can help maximize property value and improve pricing strategies. Future improvements could include incorporating additional variables such as neighborhood quality, proximity to schools, crime rates, and economic indicators to further increase prediction accuracy.